Homework 5 - Deadline December 6th

Exercise 1 ***(30 points).*** For each real number $a > 0$ such that $\cos(\ln(e^a + 1)) \neq 0$, find the equations of all the vertical and horizontal asymptotes, if they exist:

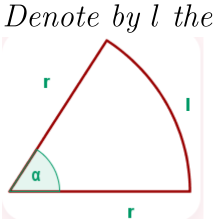
$$g(x) = \frac{\cos(x)\ln(x-a)}{\ln(e^x - e^a)}$$

Exercise 2 ***(30 points).*** Let $f(x) = \begin{cases} \sin^{2/3}(\pi x) & if \quad -1 \leq x < 2 \\ 2\tanh|x-3| & if \quad x \geq 2 \end{cases}$

(a) Find all the critical points, the intervals of increase and decrease and the local maximum and minimum values of f.

(b) Find the points of the graph where f attains the absolute maximum and minimum values.

Exercise 3 ***(20 points).*** A circular sector has a perimeter of 10 cm.
Find the radius and the central angle corresponding to the sector with the largest area, enclosed by two radii and an arc.
Denote by l the lenght of the arc, by r the radius and by α the central angle (in radians).



Exercise 4 ***(20 points).*** Evaluate the definite integral $\displaystyle\int_0^3 (x^3 + 2x)dx$ using Riemann sums.